

MATIN DAGHYANI

- ✉ matin.daghyani@gmail.com
- ✉ matin.daghyani@ece.ubc.ca
- in www.linkedin.com/in/matindaghyani

EDUCATION

University of British Columbia <i>M.A.Sc in Electrical and Computer Engineering</i> • GPA: 91.8% (A+)	Sep. 2024 – Present
Sharif University of Technology <i>B.Sc in Computer Engineering</i> • GPA: 19.22/20 • Rank: 24/199	Sep. 2019 – Feb. 2024

RESEARCH EXPERIENCE

Research Assistantship <i>University of British Columbia</i> - Under supervision of Prof. Purang Abolmaesumi - Building interpretable vision-language models for robust clinical reasoning in echocardiography	Sep. 2024 – Present
Summer Undergraduate Research Fellowship – on-site <i>Institute of Computer Science and Artificial Intelligence, INSAIT</i> - Developed a locally trained diffusion model conditioned on a single image and applied it to multi-view stereo - Investigated the use of diffusion-based generative priors for 3D reconstruction from sparse views	Jul. 2023 – Sep. 2023
Undergraduate Research Internship – remote <i>Visual Intelligence for Transportation lab (VITA), EPFL</i> - Under supervision of Prof. Alexandre Alahi - Worked on Addressing Uncertainty in Human Pose Prediction - Developed a novel method for uncertainty evaluation of human pose prediction models - Conducted literature reviews in Human Pose forecasting methods, Uncertainty Estimation methods, Graph Neural Networks, Recurrent Neural Networks, Auto Encoders, and Deep Clustering	Dec. 2021 – Aug. 2022

PUBLICATIONS

Multimodal Prototypes for Vision-Language Driven Classification of Aortic Stenosis <i>V. Wu, H. Vaseli, Matin Daghyani, B. Medhat, D. Kim, M. Y. Tsang, C. Luong, P. Abolmaesumi, T. S.M. Tsang</i>	Under Review
Toward Reliable Human Pose Forecasting with Uncertainty – pdf <i>S. Saadatnejad, M. Mirmohammadi, Matin Daghyani, P. Saremi, Y. Zoroofchi Benisi, A. Alimohammadi, Z. Tehraninasab, T. Mordan, A. Alahi</i>	IEEE RA-L journal

AWARDS AND HONORS

• Acceptance to the Summer Undergraduate Research Fellowship program at <i>INSAIT</i> Acceptance Rate: Top 1% out of over 700 applicants	Jun. 2023
• Ranked 75 in the Iran national universities entrance exam Among more than 164,000 participants	Aug. 2019

TECHNICAL SKILLS

• Programming Languages:	Python, Java, C#
• Deep Learning Frameworks:	PyTorch, TRL, Accelerate, DeepSpeed
• Version Control Systems:	Git
• Database:	MySQL, SQL Server
• Software Engineering Practices:	Clean Coding, Object-Oriented Programming, and SOLID principles
• Operating Systems:	Linux, Windows

WORK EXPERIENCE

Teaching Assistant – UBC Vancouver, Canada - Served as Teaching Assistant for three graduate-level courses in the UBC Master of Data Science program - Led in-person lab sessions, provided one-on-one student support, and contributed to course instruction and grading - Received the Best TA Award for outstanding teaching performance during the term	Jan. 2025 – Apr. 2025
Software Engineering Intern – Mohaymen ICT Company Tehran, Iran - Worked with technologies such as: C#, ASP.NET, SQL, Git, CICD - Led a team with the goal of implementing an ETL web application	Jul. 2021 – Sep. 2021
Python Instructor – Rouzbeh High School Tehran, Iran Taught basic Python programming and Algorithms to grade 10 and 11 high school students	Jan. 2021 – Jan. 2022